

1 Scope

1.1 Technology - Agentic AI

Agentic AI refers to goal-driven software entities that autonomously make decisions and take action. It refers to various architectures, techniques and frameworks for creating single agents or collaborative multiagent systems capable of unsupervised task execution. AI agents are autonomous or semiautonomous software entities that use AI techniques to perceive intents, make decisions, take actions and achieve goals.

Agentic AI entities can receive goal instructions, iterate and delegate tasks, and output variables and dynamic information — often augmenting the user’s work.

1.2 Multiagent Systems

Multiagent systems, composed of multiple, independent, interacting agents — each capable of perceiving its environment and taking actions — work toward a common goal that goes beyond the ability of individual agents.

Agentic AI will mature into fully autonomous systems where agent decisions and actions are necessary, permissible and trusted. Multiagent systems are characterized by agents imbued with:

- Access to historized views of context-based decisions and executed goals and workflows
- Dynamic adjustment of goals, decisions, plans and actions based on changing conditions

(situational awareness)

- Adaptive execution, including multiagent collaboration (such as multiagent generative systems), to learn to complete multistep processes in digital and physical environments (actuation)

1.3 Agentic Analytics

Agentic Analytics is an implementation framework for multiagent systems.

Agentic Analytics enables a shift where multistep tasks within or across workflow segments are executed autonomously, requiring only high-level user intervention. This shortens the

analytics workflow completion time, which speeds up the creation of insights that ultimately lead to faster execution and positive business impacts.

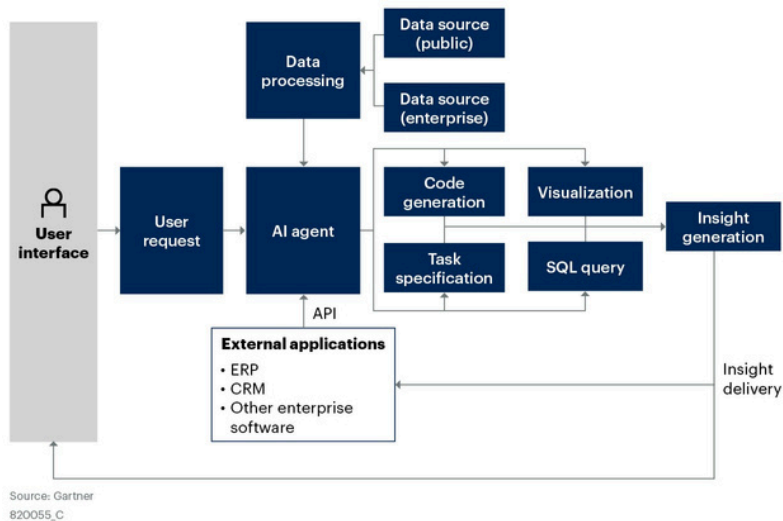


Figure 1: Building blocks of Agentic Analytics

2 Objectives

2.1 Agentic Analytics Solution

There are two vertical use cases each with its own agents / copilots. Below includes MVP definitions and data formats stacking up for each. The final web app (with mobile-friendly UI) will offer unified Agentic Analytics As a Service for these verticals’ end users.

3 Use Cases

3.1 PanoramaScore– A Flexible, Cross-Vertical Risk-Scoring Copilot

PanoramaScore is a configurable agent that ingests heterogeneous documents, enriches them through a knowledge-mesh + RAG stack, then delivers transparent quantitative risk scores for insurance, real-estate, procurement, investing, and fraud-detection workflows. It guides underwriters, analysts, and auditors with score explanations and next-step suggestions.

Five verticals are given. The goal is to pick four verticals to deliver on within the first four milestones.

3.1.1 Processing Pipeline

1. Document Ingest – User uploads file (PDF, image, CSV) via API/UI; system stores raw blob.
2. Metadata Hashing – Upload source, client ID, and checksum are written to the knowledge mesh for lineage.
3. Pre-OCR & Normalization – Images/PDF pages converted to text with Tesseract + Layout-aware OCR (e.g., LayoutLM). Format normalizers strip headers, de-skew images, redact PII, and convert everything to clean text blobs and key-value pairs.
4. NLU Copilot Extraction – Doc-type classifier, schema mapper, and transformer/R-regex extractor output structured JSON fields + confidence.
5. Vector Store & RAG – Extracted embeddings saved in pgvector; retrieval augments context with similar past cases.
6. Domain Adapter – Pluggable module maps generic fields to vertical-specific risk factors and weighting rules. That is, generic JSON →vertical features (risk factors, weights, thresholds).
7. Risk-Scoring Engine – Calibrated ensemble model generates 0-100 score, band, and factor contributions.
8. Copilot Delivery – UI/API displays score, explains drivers, and offers next-action buttons to chain downstream agents. For example:
 - Copilot receives score=78 (High), factors: “prior losses 35%, coverage gaps 22%, permit violation 11%.”
 - It highlights top spans inside the PDF and suggests actions (raise deductible; request engineering report).

3.1.2 Domain-Adapter Schematic (PanoramaScore v1)

Adapters are hot-swappable YAML files containing weight matrices, feature transforms, and threshold tables—allowing rapid expansion to new industries without code rewrites.

Vertical	Inputs	Primary Key Fields Extracted	Core Risk Factors	Score Bands (0-100)	Key Regs to Watch
Insurance	Application forms, prior claims, credit scores, lifestyle data, ISO claim histories	Claim frequency, severity, exposure type, credit-risk proxy, lifestyle/risk behaviors	Prior claims %, unreported gaps, high liability classes, coverage gaps, suspicious claim patterns	0-39: High Risk, 40-69: Moderate Risk, 70-100: Preferred Risk	State DOI ratefiling, FCRA, NAIC modelregs
Supply Chain	Shipment records, supplier financials, dispute feeds, ERP logs	Supplier tier, delivery latency, financial fragility, geo-political flags	Data sparsity, inconsistent formats, vendor cooperation	0-39: High Risk, 40-69: Moderate Risk, 70-100: Low Risk	None early; voluntary ESG/SEC guidelines (U.S.)
Cross-Asset Investing	Property appraisals, financials, sector data, market liquidity history	ROI %, rehab cost, sector beta, liquidity ratio, yield-to-risk	Illiquid data, inconsistent reporting, macro correlation risk, liquidity gap	0-39: Reject, 40-74: Opportunistic, 75-100: Core	SEC Reg D, FINRA 2210 (advisory marketing)
Fraud Detection	Transaction logs, behavioral analytics, device IDs, chargeback history	Velocity, geolocation mismatch, unusual routing, historical fraud flags, geo-IP	Adversarial drift, real-time latency, geo mismatch, duplicate payees	0-29: Likely Fraud, 30-69: Suspicious, 70-100: Clean	PCI-DSS, AML/CFT, SOC2
Real Estate	MLS listings, public records, flood/fire risk maps, zoning data	Build year, land use code, historical sale delta, infrastructure score	Regional data variation, outdated records, zoning opacity	0-39: High Local Risk, 40-69: Value-Add, 70-100: Core	zoning laws, Fair Housing Act

3.1.3 MVP Definition of Done

1. End-to-End Pipeline converts uploaded PDFs (insurance + procurement pilots) into risk scores in < 20s.
2. Extraction Accuracy $\geq 90\%$ F1 on critical fields (claim counts, DUNS score, AVM).
3. Score Calibration K-S statistic < 0.1 against pilot holdout set.
4. UI Widget visualizes factor contributions and offers action buttons that call downstream agents.
5. Audit Trail Export delivers JSON containing original spans, mesh hashes, and model version.

3.1.4 KPI Metrics

- 1 Average Turnaround Time $\leq 20s$ per document (95th pct).
- . AutomationRate $\geq 50\%$ of decisions closed without human override by month 6.
- 2 Extraction Drift $\leq 3\%$ high-severity drift incidents per quarter.
- . Client ROI documentable 30% reduction in manual review cost.

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3.1.5 User Flow Archetype

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1. Step 1: Upload
 - .
 - User uploads a policy file, claim sheet, vendor contract, or billing form via UI or API.
 - System parses document → hashes metadata → stores in mesh.
2. Step 2: Copilot Trigger
 - User selects “Run Risk Check” from Copilot dashboard.
 - Document is classified and aligned with the correct domain adapter.
3. Step 3: Score Generation
 - Vector-based RAG fetches context.
 - Copilot interprets document and context → runs scoring agent.
 - Risk score with contributing factors is generated in < 10s.

4. Step 4: Review & Action

- Score displayed with label (e.g., "Moderate Risk").
- User can: Accept, Override, or Request Deeper Explanation.
- Action is logged in the audit trail with time and user ID.

For example:

1. Commercial Insurance Broker

- Upload loss run + ISO report.
- Score = 12 (High Risk) due to seven losses and gap in prior coverage.
- Copilot recommends raising deductible and scheduling a loss-control survey.
- Broker forwards score + suggestions to underwriter via in-app chat, attaching RAG-pulled benchmark of similar insureds.

3.2 InvestorLink Copilot

A mobile-first agentic copilot that matches investors with real-estate fix-and-flip projects and live stock/ETF plays. The Copilot cross-references MLS data, Redfin/Zillow APIs, contractor portfolios, and real-time market feeds (Yahoo Finance, Alpaca) to surface blended opportunities (“refinance flip, hedge with S&P puts”).

In other words, it’s an app that matches: Investors → Realtors → Homeowners → Contractors for fixer-uppers and connects: Investors → Broker-dealers for equities/ETFs. In the former, Investors browse opportunities; Copilot projects returns. Realtors & Homeowners list properties; Copilot prices them dynamically.

3.2.1 Agentic Features

- Recommends property flips based on local comps, permit activity, and ROI projections
- Suggests stock-index fund pairings based on user risk profile and real estate exposure
- Facilitates in-app chat and deal flow between users, brokers, and service pros
- Autonomous agent handles risk balancing and timing advice

3.2.2 Core Agents

- FlipScout: screens MLS + off-market listings, scores ROI, auto-pings local contractors.
- TradeSignal: ingests public market APIs, suggests stock/ETF re-balances to fund projects.
- DealDesk: auto-drafts purchase offers, rehab budgets, and brokerage orders.

3.2.3 MVP Definition of Done

1. Property Match Agent: Identifies high-ROI fixer-upper listings based on investor profile and local market data.
2. Contractor Connect: Automatically pairs selected properties with verified local contractors for instant rehab quotes.
3. Broker Bridge: Links investors to registered stock brokers for portfolio reallocation aligned with property funding needs.
4. ROI Projection Engine: Generates investment forecast reports combining real estate appreciation and equity market performance.
5. Offer & Order Flow: Auto-generates home purchase offers and stock trade orders through guided investor Copilot flows.

3.2.4 Data Sources for Agentic Analytics

1. MLS & Real Estate APIs (Sources: Zillow, Redfin, Realtor.com): Provides property listings, price history, location data, and property condition for investment scoring.
2. Financial Market APIs (Sources: Yahoo Finance, Alpha Vantage, FRED): Offers real-time stock/ETF prices, market trends, and economic indicators for portfolio analysis and funding decisions.
3. Public Contractor & Permitting Data (Sources: City permitting databases, Yelp, Angi API): Supplies contractor reputations, licensing info, project history, and average rehab timelines/costs for risk-adjusted planning.

3.2.5 KPI Metrics

1. Match-to-Engagement Rate: Percentage of property-investor matches that lead to active contractor or broker engagement. (Target: 30%+)

2. Projected ROI Accuracy: Variance between forecasted and actual ROI on completed deals. (Target: <10% deviation)
3. Investor Activation Rate: Proportion of registered investors who submit at least one offer or brokerage order. (Target: 30% in 30 days)
4. Time-to-Offer: Average time from property match to investor submitting an offer. (Target: <72 hours)

4 User Interface of Copilots

- Context-Aware Sidebar: A floating Copilot panel that provides real-time suggestions tied to the user's current screen or workflow.
- Copilot AI Effects: Include a hovering or pulsing AI orb that subtly animates when providing suggestions—think Siri meets Bloomberg Terminal. Integrate typing trails, decision chains, and task progress ripples to show how the agentic AI is thinking and acting. For actions like portfolio updates, use dynamic overlays that simulate intelligent agent routing in real time.
- Adaptive Templates: Dynamically update UI components—such as forms—based on prior user inputs and historical data.
- Role-Based Access Control: Enforce user permissions within the Copilot, ensuring sensitive data is only visible to authorized roles.
- Secure Data Flow: Employ end-to-end encryption, logging, and multi-factor authentication to protect in-app communications and document exchanges.
- Bot Protection: Integrate reCAPTCHA at signup to ensure authentic user registrations and prevent bot spam.
- Smart Throttling: Display friendly UI cooldown timers after excessive requests to prevent system overload gracefully.

5 KPIs

The defensible KPIs for the Agentic Analytics Solution are

1. Fault Tolerance (EDA): Agents operate on an event-driven architecture, allowing them to respond to real-time triggers. Event brokers can manage asynchronous communication between agents, ensuring agents process tasks efficiently and maintain responsiveness under high loads.

2. Unified Data Access: Develop a flexible data access layer using RESTful or GraphQL APIs to unify structured (SQL/CSV) and unstructured (DOC, DOCX, PDF, TXT, JSON) data. Leverage a modular pipeline design with schema adapters per vertical.

3. PanoramaScore

- ☐ Processing Pipeline fully designed, tested and delivered
- ☐ MVP Definition of Done ticked off
- ☐ KPIs metrics met

4. InvestorLink

- ☐ Core agents built out
- ☐ MVP Definition of Done ticked off
- ☐ KPIs metrics met

5. Agent-Oriented Automation

- ☐ Use LangChain or AutoGen to orchestrate task-based AI agents (e.g., Fetcher Agent, Parser Agent, Validator Agent)
- ☐ Create a Copilot Interface to allow user interaction and feedback on multi-step agent workflows
- ☐ Build intent routing logic that dynamically assigns the right agents based on vertical, task, and user context
- ☐ Explainability Layer – SHAP-style bar chart plus original text spans satisfy regulatory transparency

6 Prelaunch Necessities

Launch a prelaunch beta signup landing page, clearly highlighting early-bird benefits, such as free premium access for the first month and exclusive early features. The goal is to drive traffic through targeted LinkedIn ads and email outreach and have 50+ qualified beta user signups within 4 weeks. Monitor metrics such as signup conversion rates (targeting 20%+), landing page traffic, and campaign cost-per-signup. Regularly engage the beta list via email updates to maximize launch readiness and adoption.

7 Credit-Based Subscription Model

This is a structured monetization framework for the solution using Pay-As-You-Go + Lapis Credits + Subscription Tiering, differentiated by verticals (Supply Chain, Insurance, Investing, Fraud Detection, and Real Estate) based on adapter complexity and model cost. Users can unlock premium features and agentic tasks anytime by purchasing additional lapis credits, ensuring flexibility without upgrading their subscription tier.

- Transparent Costing: Each task clearly shows the lapis it consumes before execution.
- Instant Top-Up Options: Buy lapis bundles and continue workflows without disruption.

7.1 PanoramaScore

7.1.1 Pay-As-You-Go Pricing (At the End of Onboarding Phase)

Vertical	Minimum Charge to Activate Use	PerTask Charge (Lapis)	Adapter Complexity
Supply Chain	\$50	8–15 Lapis	Medium
Insurance	\$100	15–25 Lapis	High
Investing	\$75	12–20 Lapis	Medium
Fraud Detection	\$60	10–18 Lapis	Medium
Real Estate	\$40	6–12 Lapis	Medium

7.1.2 On-Dashboard Experience (Post-Onboarding)

- Transparent CostingEach task clearly displays the number of Lapis credits it will consume before execution.
- Instant Top-Up Options: Users can purchase Lapis bundles as needed:
 - 50 Lapis – \$40
 - 100 Lapis – \$75
 - 250 Lapis – \$175

7.1.3 Subscription Option (Post-Onboarding Switch)

Once initial onboarding Lapis credits are used, users may switch to a Tier customized per vertical. Each tier includes:

- A set number of Lapis credits
- Reserved execution priority
- Historical logs & audit trail
- API access for data pulls (where available)

Monthly Subscription

Vertical	Tier Name	Monthly Price	Included Lapis	Special Feature
Supply Chain	LapisFlow	\$129	150	Vendor risk scoring history + batch file upload
Insurance	LapisShield	\$199	200	Scenario simulation + regulatory form generator
Investing	LapisEdge	\$179	175	Dynamic ROI recalibration + risk-adjusted ranking
Fraud Detection	LapisSentinel	\$149	160	Real-time API risk pings + velocity tuning
Real Estate	LapisCore	\$109	140	Property profile builder + market overlay reports

7.1.4 Free Trial Logic by Vertical

Each vertical includes a free trial period based on model complexity and expected evaluation time. The trial gives the user a starter pack of Lapis credits (non-renewable unless converted to paid tier).

Vertical	Free Trial Duration	Free Trial Lapis Credits	Trial Features
Supply Chain	5 days	40	Up to 5 vendor profiles, latency dashboards, basic disruption flagging
Insurance	5 days	30	Underwriting simulation, claim pattern insights, rule-based red flag preview
Investing	6 days	35	3 asset deals, rehab cost analyzer, ROI projection sandbox
Fraud Detection	4 days	30	Velocity logic test, anomaly explorer, device fingerprint simulator
Real Estate	6 days	45	5 property scoring runs, zoning sensitivity overlay, valuation spread estimator

7.2 InvestorLink

Pay-As-You-Go pricing model for InvestorLink is structured around its core modules—FlipScout, TradeSignal, and DealDesk—with Lapis-based billing, free trial logic, transparent costing, and top-up/subscription options.

7.2.1 Pay-As-You-Go Pricing (At the End of Onboarding Phase)

Module	Features	Per Task Charge (Lapis) 10–18	Minimum Model Charge to Activate Use	Cost
FlipScout	Screens MLS + off-market listings, scores ROI, auto-pings local contractors		\$75	Medium-High
TradeSignal	Ingests public market APIs, suggests stock/ETF rebalances to fund property projects	12–20	\$75	High
DealDesk	Auto-drafts purchase offers, rehab budgets, and brokerage orders	8–15	\$75	Medium

7.2.2 On-Dashboard Features (Post-Onboarding)

- Transparent Costing: Every action clearly shows how many Lapis it will consume before execution.
- Instant Top-Up Options: Users can purchase additional Lapis bundles:
 - 50 Lapis – \$40
 - 100 Lapis – \$75
 - 250 Lapis – \$175

7.2.3 Monthly Subscription – LapisEdge Plan (Post-Onboarding Switch)

Available once onboarding trial ends or Lapis credits are exhausted.

Tier Name	Monthly Price	Included Lapis Credits	Features
LapisEdge	\$179	175	Unlimited FlipScout runs, scheduled TradeSignal alerts, full DealDesk automation

7.2.4 Free Trial Logic

Each vertical includes a free trial period based on model complexity and expected evaluation time. The trial gives the user a starter pack of Lapis credits (non-renewable unless converted to paid tier).

Free Trial Duration	Free Trial Lapis Credits	Trial Features
5 Days	40 Lapis	Run 1 full FlipScout scans, 1 TradeSignal analysis, and 1 DealDesk offer/budget draft

Project	Timeline	Retrospective
AgenticAnalytics Solution	10-12wks	1wk

For specific milestones, see below. The deliverables for each milestone appear thereafter.

Milestone	Timeline	PayPercent
PanoramaScore Processing Pipeline →1st Vertical	wks1,2	15
PanoramaScore2nd Vertical	wks3,4	1
PanoramaScore3rd Vertical	wks5,6	5
PanoramaScore4th Vertical & QA & Deployment	wks7,8	1
InvestorLinkTwoof Core Agents (3.2.2)		5
InvestorLink Completion & Final Deployment	wks9,10 wks11,12	2 0

8.2 Deliverables

The Offer is contingent upon successful release of each milestone's deliverables. A 12-week delivery plan is concurred upon with bi-weekly agile sprints. The list of deliverables will be conferred once the proposal has been submitted.

- Weeks 1-2: Full backend & frontend stable version of PanoramaScore Processing Pipeline applied to the first vertical and delivered to the Client.
- Weeks 3-4: As defined in the above table.
- Weeks 5-6: As defined in the above table.
- Weeks 7-8: As defined in the above table.
- Week 9-10: As defined in the above table.
- Week 11-12: As defined in the above table.

9. Offer's Terms

All milestone deliverables, including but not limited to code, documents, and any other project artifacts, are released exclusively to the Client. You agree that all intellectual property created under this project is the sole property of the Client. No sharing, reuse, distribution, or disclosure of any intellectual property or project-related materials to any third party is permitted without the Client's explicit written consent.

The first milestone serves as a proof of concept to confirm mutual agreement between You and the Client before proceeding further. If a milestone deliverable fails to meet the agreed-upon quality standards, the Client reserves the right to terminate the contract. As a courtesy, You will be given three (3) working days to address and correct the deficiencies. Should You fail to satisfactorily remedy the issues, the Client will proceed under Upwork's policies for fixed-price contracts to request a refund of the escrowed funds. Both parties acknowledge that Upwork's decision regarding fixed-price milestone disputes will be final, and agree to part ways professionally and tactfully if this process is invoked.